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Department of Information Technology

Assignment

Date: 22/04/2025

Subject: Web Application Development (2024-25)

Q. No.	Question
Q.1)	Illustrate the need Through Package
Q.2)	Implement small application which will make use of mentioned package (I/p, O/p)
Q.3)	Illustrate the need for code of ethics

Q.1) Illustrate the need Through Package

Ans:

The "Through" package is a small Node.js module that helps in working with streams—a way to handle continuous flow of data. In web development, we often deal with files, user input, APIs, or live data that comes in small parts. Using "Through", we can create a Transform Stream that reads this data, processes it (like converting or filtering), and then passes it ahead.

Needs:

- It allows real-time data processing.
- Reduces the need for temporary storage.
- Helps keep code short, clean, and easy.
- Makes it easier to modify, filter, or change data as it passes *through* the application.

Where to use:

- In real-time apps like chats or data dashboards.
- In backend systems that handle file uploads/downloads.
- In data logging and reporting systems.
- In building APIs that modify data before sending it to the user.

Real-life examples :

- Converting a large text file to uppercase while reading it.
- Filtering out sensitive words from chat messages.
- Modifying or encrypting user input before saving it.

Without "Through", writing such logic would require more code and can be confusing. This package simplifies that work and is helpful in backend or full-stack projects where data needs to be handled efficiently.

Q.2) Implement small application which will make use of mentioned package (I/p, O/p)

Ans:

Code:

```
const through = require('through');
const fs = require('fs');

// Create a transform stream that converts data to uppercase
const toUpperCase = through(function (chunk) {
  // Convert and pass the data
  this.queue(chunk.toString().toUpperCase());
});

// Read from input.txt, pass through transformer, and write to output.txt
fs.createReadStream('input.txt')
  .pipe(toUpperCase)
  .pipe(fs.createWriteStream('output.txt'));
console.log("Text converted successfully.");
```

Input File (input.txt):

```
hello world  
this is a test file  
made for demo purpose
```

Output File (output.txt):

```
HELLO WORLD  
THIS IS A TEST FILE  
MADE FOR DEMO PURPOSE
```

Q.3) Illustrate the need for code of ethics

Ans:

A Code of Ethics is a set of rules or principles that developers follow to make sure their work is honest, fair, and respectful. In the world of web development, we deal with people's data, privacy, and digital life—so being ethical is very important.

Need of code of ethics:

1. To protect privacy – Users share their personal data with our websites. We must protect it.
2. To prevent misuse – Developers should not use tools or code to spy, hack, or cheat.
3. To be transparent – Always tell users what your application does with their data.
4. To ensure fairness – Applications should work equally well for all users without bias.
5. To follow laws – Developers must respect legal and copyright rules.

Code of Ethics in "Through":

Using the "Through" package to modify user messages in a chat app. An ethical developer will:

- Inform users that their messages are being filtered or processed.
- Avoid collecting or storing messages without permission.
- Never change the meaning of a user's message.
- Keep logs secure and private.

By following a code of ethics, we can build trustworthy and safe applications that people can depend on. It also helps us become responsible professionals in the tech world.